

UST-TSILMA, Russian Federation

WMO: 234050

Lat: 65.4387N

Lon: 52.1974E

Elev: 79

StdP: 100.38

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.6	(c) -33.3	(d) -39.4	(e) 0.1	(f) -36.5	(g) -36.1	(h) 0.1	(i) -33.2	(j) 9.4	(k) -11.7	(l) 8.7	(m) -11.7	(n) 2.0	(o) 290	(p) 0.724

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	26.8	19.0	24.5	17.7	22.3	16.4	20.1	25.2	18.7	23.1	17.2	21.1	2.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.2	22.6	16.8	12.1	21.1	15.4	11.1	19.7	57.8	25.4	53.3	23.2	48.8	21.1	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.7	7.6	6.8		-37.8	29.4	4.5	2.1	-41.1	31.0	-43.7	32.2	-46.2	33.4	-49.5	35.0	
				DB												
				WB	-37.8	21.5	4.3	1.7	-41.0	22.7	-43.5	23.6	-45.9	24.6	-49.1	25.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.8	-16.3	-15.6	-8.1	-1.5	4.9	11.5	15.4	12.0	7.2	0.4	-8.2	-12.5		
	DBStd	12.56	9.06	9.16	6.70	5.67	6.04	5.33	4.89	4.03	3.63	4.45	7.68	9.02		
	HDD10.0	4338	816	716	561	346	183	46	8	23	98	299	545	697		
	HDD18.3	7028	1074	949	819	595	416	214	119	200	334	557	795	955		
	CDD10.0	394	0	0	0	1	26	90	175	86	15	1	0	0		
	CDD18.3	41	0	0	0	0	1	9	27	4	0	0	0	0		
	CDH23.3	309	0	0	0	0	9	69	202	28	1	0	0	0		
	CDH26.7	58	0	0	0	0	1	12	42	4	0	0	0	0		
(14) Wind	WSAvg	3.4	3.5	3.5	3.6	3.6	3.7	3.5	3.1	3.1	3.1	3.4	3.3	3.4		
(15) Precipitation	PrecAvg	550	33	27	28	31	43	56	64	67	58	61	43	40		
	PrecMax	769	61	68	83	87	79	135	144	143	116	120	94	82		
	PrecMin	401	17	3	4	8	9	7	30	8	17	17	18	8		
	PrecStd	91	11	12	15	17	17	28	25	28	23	23	16	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.3	1.8	6.4	15.6	24.5	28.5	30.2	26.5	20.2	12.4	4.9	1.7		
		MCWB	-0.5	1.0	3.0	9.1	15.2	18.8	20.6	19.6	14.8	9.5	4.1	0.7		
	2%	DB	-1.4	0.0	2.9	10.1	21.1	25.3	27.8	23.1	16.6	9.5	2.8	0.7		
		MCWB	-2.0	-0.8	0.7	5.4	13.0	17.4	19.8	17.7	12.8	7.9	2.0	0.1		
	5%	DB	-2.8	-1.7	1.6	7.4	18.0	22.7	25.8	20.8	14.5	7.6	1.5	-0.1		
		MCWB	-3.4	-2.4	0.0	3.5	11.4	15.6	18.9	16.3	11.9	6.5	0.8	-0.7		
	10%	DB	-4.6	-3.4	0.3	5.5	14.9	20.2	23.5	18.7	12.6	6.2	0.5	-1.5		
		MCWB	-5.2	-4.1	-0.9	2.6	9.6	14.3	17.6	15.4	10.6	5.2	-0.1	-2.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.4	1.0	3.2	9.3	16.3	20.1	22.2	20.5	15.3	10.1	4.1	1.1		
		MCDB	0.1	1.7	5.8	14.6	22.7	26.2	27.7	25.2	19.1	11.5	4.7	1.5		
	2%	WB	-1.9	-0.8	1.2	5.8	14.0	18.2	20.9	18.7	13.5	8.0	2.2	0.2		
		MCDB	-1.5	-0.2	2.4	9.5	20.1	23.9	26.3	21.6	15.9	9.5	2.6	0.7		
	5%	WB	-3.4	-2.4	0.3	4.2	12.0	16.7	19.6	17.0	12.1	6.6	1.0	-0.8		
		MCDB	-2.8	-1.7	1.3	6.6	16.9	21.3	24.5	19.7	14.1	7.6	1.4	-0.2		
	10%	WB	-5.2	-4.1	-0.9	2.9	9.8	14.9	18.3	15.6	10.8	5.3	0.0	-2.1		
		MCDB	-4.6	-3.5	0.4	5.1	14.5	19.4	22.7	18.3	12.4	6.2	0.4	-1.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	6.3	7.8	7.8	8.5	8.4	8.6	7.4	6.0	3.8	4.8	5.7		
		MCDBR	5.9	5.4	6.3	9.8	12.8	11.6	11.3	11.2	9.4	5.0	3.8	4.7		
	5% WB	MCWBR	5.8	4.9	5.0	6.5	7.2	5.9	5.6	6.2	5.9	4.2	3.8	4.6		
		MCWBR	6.0	5.4	5.6	8.3	11.9	10.7	10.4	9.8	8.5	4.9	3.7	4.7		
(40) Clear-Sky Solar Irradiance	taub		0.216	0.270	0.305	0.338	0.344	0.354	0.411	0.371	0.329	0.292	0.214	0.214		
		taud	1.912	2.057	2.103	2.149	2.393	2.464	2.297	2.394	2.492	2.304	1.960	1.923		
	Ebn at Noon		243	587	749	818	861	859	785	787	749	579	238	44		
		Edh at Noon	22	60	93	114	100	96	110	91	66	48	19	6		
(44) All-Sky Solar Radiation	RadAvg		0.12	0.74	2.21	3.77	4.94	5.09	4.99	3.30	1.73	0.62	0.16	0.03		
	RadStd		0.01	0.09	0.26	0.41	0.41	0.57	0.54	0.35	0.18	0.08	0.03	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page